



TMA4180

Vectors on Machine–Support

Group Project Report

Solving the Soft-Margin SVM: Theory and Implementation

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Semester:

Spring 2025

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April 30, 2025

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Notation

Symbol	Description
$\mathbf{w} \in \mathbb{R}^d$	Weight vector
$b \in \mathbb{R}$	Bias term
$\boldsymbol{\xi} \in \mathbb{R}^M$	Vector of slack variables
$X = [\mathbf{x}_1, \dots, \mathbf{x}_M] \in \mathbb{R}^{d \times M}$	Data matrix
$G = X^\top X \in \mathbb{R}^{M \times M}$	Gram matrix
$\mathbf{y} = [y_1, \dots, y_M]^\top \in \{\pm 1\}^M$	Label vector
$Y = \text{diag}(y_1, \dots, y_M) \in \mathbb{R}^{M \times M}$	Label/diagonal matrix
$C > 0$	Penalty parameter for misclassification

Table 1: Notation for the Soft-Margin classification problem

1 Introduction

Support Vector Machines (SVMs) are a theoretically grounded method for binary classification, particularly effective in high-dimensional settings. Given a set of labelled training data, the aim is to construct a decision function that separates the two classes with maximal margin while tolerating potential misclassifications. This leads to the formulation of the soft-margin SVM, a convex optimization problem balancing model complexity and classification error via a regularization parameter.

A theoretical and numerical investigation of the soft-margin SVM formulation is the goal of this report. The existence and uniqueness of solutions is discussed through exploration of the structure and behavior of the optimization landscape using convex analysis. A numerical method is implemented for solving soft-margin SVMs, and tested on numerical experiments, demonstrating the performance of the algorithm.

2 Problem Formulations

2.1 The Primal Problem (P)

We begin by formulating the Soft-Margin classification as a constrained optimization problem, which will be referred to as the *Primal Problem* (P).

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} f(\mathbf{w}, b, \boldsymbol{\xi}) = \frac{1}{2} \|\mathbf{w}\|_2^2 + C \mathbf{1}_M^\top \boldsymbol{\xi} \quad \text{subject to} \quad \begin{cases} c_1(\mathbf{w}, b, \boldsymbol{\xi}) = \mathbf{1}_M - Y(X\mathbf{w} + b\mathbf{1}_M) - \boldsymbol{\xi} \leq \mathbf{0}_M \\ c_2(\boldsymbol{\xi}) = -\boldsymbol{\xi} \leq \mathbf{0}_M \end{cases} \quad (\text{P})$$

2.1.1 Existence and Uniqueness of Constrained Optimization Problems

For an unconstrained optimization problem, three properties are required to prove the existence and uniqueness of a minimizer; *Feasibility*, *Coercivity*, and *Strict Convexity*. Feasibility guarantees a non-empty constraint set, coercivity prevents minimizers from escaping to infinity, and strict convexity ensures both global optimality and uniqueness. Together these properties ensure that a unique global minimizer exists.

However, for constrained optimization, these traits do not by themselves guarantee existence or uniqueness of the minimizer, because the structure of the feasible set can allow none or multiple solutions even when the objective is strictly convex. Constraints can disconnect or restrict the domain in ways that create no or multiple global minimizers despite the objective's properties.

Theorem 1. Existence and Uniqueness of the Primal Solution

The primal problem (P) always admits at least one global minimizer, given that the feasible set is non-empty, closed and bounded, and the objective function is continuous. Additionally, if the feasible set is convex and the objective is strictly convex in $\mathbf{w}$, then the optimal weight vector $\mathbf{w}^\star$ is unique. However, the optimal bias $b^\star$ and slack variables $\boldsymbol{\xi}^\star$ may not be uniquely determined in general.

Proof of Theorem 1

Existence of a minimizer: To show the existence of a minimizer in a constrained optimization problem, the objective function must be continuous and the feasible set must be non-empty and closed. Additionally, either the feasible set must be bounded or the objective function must be coercive in order to guarantee that the minimizer does not escape to infinity, effectively bounding the feasible set.

From the constraints given in (P), the feasible set is defined by two sets of affine inequalities:

$$\Omega = \{(\mathbf{w}, b, \boldsymbol{\xi}) \in \mathbb{R}^{d+1+M} \mid Y(X\mathbf{w} + b\mathbf{1}_M) + \boldsymbol{\xi} \geq \mathbf{1}_M, \boldsymbol{\xi} \geq \mathbf{0}_M\}$$

- **Continuity:** The objective function $f(\mathbf{w}, b, \boldsymbol{\xi})$ is continuous, as it is the sum of continuous functions.
- **Feasibility:** Setting $\mathbf{w} = \mathbf{0}_d$, $b = 0$, and $\boldsymbol{\xi} = \mathbf{1}_M$ satisfies all constraints. Thus, the feasible set is nonempty, $\Omega \neq \emptyset$.
- **Closedness:** The feasible set is defined by continuous affine inequality constraints, making it closed. For any convergent sequence (x_k) in the feasible set with limit x , the continuity of the constraint functions ensures $g_i(x) = \lim_{k \rightarrow \infty} g_i(x_k) \leq 0$ for all constraints, thus x remains feasible and the set is closed.
- **Boundedness:** Although the feasible set may not be bounded, the objective function is coercive, as it grows unboundedly $\lim_{\|\mathbf{w}\|, \|b\|, \|\boldsymbol{\xi}\| \rightarrow \infty} f(\mathbf{w}, b, \boldsymbol{\xi}) \rightarrow \infty$ with $C > 0$ and $\xi_i \geq 0$. Thus, coercivity prevents minimizers from escaping to infinity, ensuring they lie in a bounded region.

By the *Extreme Value Theorem*, since the feasible set is non-empty and closed, and since coercivity prevents escape to infinity while the objective is continuous, a global minimizer must exist[Roc70].

Uniqueness of a minimizer: To show the uniqueness of a minimizer in a constrained optimization problem, the objective function must be strictly convex, and the feasible set must be convex. Strict convexity ensures that the objective cannot have two distinct minimizers achieving the same value, while convexity of the feasible set guarantees that any convex combination of feasible points remains feasible. Together, these properties imply that if a minimizer exists, it must be unique.

- **Strict Convexity:** The objective function is strictly convex in $\mathbf{w}$, since it includes the term $\frac{1}{2}\|\mathbf{w}\|_2^2$, which is strictly convex. The dependence on b and $\boldsymbol{\xi}$ is affine, which does not affect strict convexity in $\mathbf{w}$.
- **Convexity of the Feasible Set:** The feasible set is defined by affine inequalities, and sets defined by affine inequalities are convex. Hence, the feasible region is convex.
- **Behavior of b and $\boldsymbol{\xi}$:** Since b and $\boldsymbol{\xi}$ appear linearly in the objective and constraints, and because strict convexity is only in $\mathbf{w}$, the variables b^* and $\boldsymbol{\xi}^*$ may not be uniquely determined, especially in degenerate cases.

Thus, by strict convexity of the objective in $\mathbf{w}$ and convexity of the feasible set, the minimizer $\mathbf{w}^*$ is unique. Furthermore, since the problem is convex, any local minimizer is also a global minimizer. $\square$

2.2 The Dual Problem (DP)

The primal problem involves minimizing a regularized hinge loss over three sets of variables: the weight vector $\mathbf{w} \in \mathbb{R}^d$, the bias $b \in \mathbb{R}$, and the slack variables $\boldsymbol{\xi} \in \mathbb{R}^M$. Solving this problem directly can be computationally inefficient, especially in high-dimensional settings. An alternative formulation, derived via Lagrangian duality, reformulates the problem in terms of a single dual variable $\boldsymbol{\alpha} \in \mathbb{R}^M$, which provides both computational and theoretical advantages. This alternative formulation, shown below, will be referred to as the dual problem (DP).

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^M} \tilde{f}(\boldsymbol{\alpha}) = \frac{1}{2} \langle \boldsymbol{\alpha}, \mathbf{YGY}\boldsymbol{\alpha} \rangle - \langle \mathbf{1}_M, \boldsymbol{\alpha} \rangle \quad \text{subject to} \quad \begin{cases} 0 \leq \boldsymbol{\alpha} \leq C\mathbf{1}_M, \\ \boldsymbol{\alpha}^\top \mathbf{Y} = 0. \end{cases} \quad (\text{DP})$$

2.2.1 Dual Derivation

The dual formulation offers several key advantages. In the context of support vector machines, solving the dual allows the optimization to depend only on the number of data points rather than the dimension of the spatial space, making it efficient in high-dimensional settings. Additionally, the dual formulation naturally incorporates the kernel trick, enabling non-linear decision boundaries without explicitly mapping data to higher dimensions. It also gives insight into which data points are support vectors, since only the corresponding dual variables are non-zero. Thus, the dual problem is usually more numerically stable and computationally efficient, than the primal formulation.

Theorem 2. Dual Problem Derivation

The dual problem (DP) is derived from the primal problem (P) using Lagrangian multipliers and KKT conditions.

Proof of Theorem 2

Lagrangian To derive the dual problem, we start with the primal problem (P). We introduce Lagrange multipliers $\alpha_1, \alpha_2 \geq 0$ for the constraints $c_1(\mathbf{w}, b, \boldsymbol{\xi}) \leq \mathbf{0}_M$ and $c_2(\boldsymbol{\xi}) \leq \mathbf{0}_M$, respectively. The Lagrangian is given by:

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \alpha_1, \alpha_2) = \frac{1}{2} \|\mathbf{w}\|_2^2 + C\mathbf{1}_M^\top \boldsymbol{\xi} + \alpha_1^\top c_1(\mathbf{w}, b, \boldsymbol{\xi}) + \alpha_2^\top c_2(\boldsymbol{\xi}).$$

Because $\mathcal{L}$ is convex in $(\mathbf{w}, b, \boldsymbol{\xi})$ and concave in (α_1, α_2) , the minimax theorem [Neu28] implies

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \max_{\alpha_1, \alpha_2} \mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \alpha_1, \alpha_2) = \max_{\alpha_1, \alpha_2} \min_{\mathbf{w}, b, \boldsymbol{\xi}} \mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \alpha_1, \alpha_2) = \max_{\alpha_1, \alpha_2} g(\alpha_1, \alpha_2)$$

KKT conditions To find the minimum of the Lagrangian, we need to satisfy the KKT conditions. The KKT conditions are a set of necessary and sufficient conditions for optimality in constrained optimization problems. They consist of three components:

Stationarity: The gradients of the Lagrangian with respect to the primal variables $(\mathbf{w}, b, \xi)$ must vanish:

$$\begin{aligned}\nabla_{\mathbf{w}} \mathcal{L} &= \mathbf{w} - X^T Y \alpha_1 = \mathbf{0}_M, \\ \nabla_b \mathcal{L} &= -\alpha_1^T Y = \mathbf{0}_M^T, \\ \nabla_{\xi} \mathcal{L} &= C \mathbf{1}_M^T - \alpha_1^T - \alpha_2^T = \mathbf{0}_M^T.\end{aligned}$$

Complementary slackness: The Lagrange multipliers α_1 and α_2 must satisfy the complementary slackness conditions:

$$\begin{aligned}\alpha_1^T c_1(\mathbf{w}, b, \xi) &= \alpha_1^T (\mathbf{1}_M - Y(X\mathbf{w} + b\mathbf{1}_M) - \xi) = \mathbf{0}_M^T, \\ \alpha_2^T c_2(\xi) &= -\alpha_2^T \xi = \mathbf{0}_M^T.\end{aligned}$$

Feasibility: The primal variables $(\mathbf{w}, b, \xi)$ must satisfy the original constraints:

$$\begin{aligned}c_1(\mathbf{w}, b, \xi) &= \mathbf{1}_M - Y(X\mathbf{w} + b\mathbf{1}_M) - \xi \leq \mathbf{0}_M, \\ c_2(\xi) &= -\xi \leq \mathbf{0}_M.\end{aligned}$$

Thus, we get the following KKT conditions:

$$\begin{aligned}\mathbf{w} &= X^T Y \alpha_1, \\ \alpha_1^T Y &= \mathbf{0}_M^T, \\ \mathbf{0}_M &\leq \alpha_1 \leq C \mathbf{1}_M.\end{aligned}$$

Where $\alpha_2 = \mathbf{0}_M$ because of the complementary slackness condition. Substituting this in to the Lagrangian, denoting $\alpha = \alpha_1$ and flipping signs for a minimization problem, we get:

$$\min_{\alpha} \tilde{f}(\alpha) := \frac{1}{2} \langle \alpha, YGY\alpha \rangle - \langle \mathbf{1}_M, \alpha \rangle \quad \text{subject to} \quad \begin{cases} 0 \leq \alpha \leq C \mathbf{1}_M, \\ \alpha_1^T Y = 0. \end{cases}$$

This completes the derivation of the dual problem (DP). □

Once the dual problem (2) has been solved and an optimal dual variable α^* has been obtained, we can reconstruct the corresponding solution $(\mathbf{w}^*, b^*, \xi^*)$ to the primal problem (P) using the KKT optimality conditions.

Proposition 1. Recovering Primal Solution from Dual Solution

Given an optimal solution α^* to the dual problem (DP), the corresponding optimal solution $(\mathbf{w}^*, b^*, \xi^*)$ to the primal problem (P) can be recovered as:

$$\begin{aligned}\mathbf{w}^* &= X^T Y \alpha^* \\ b^* &= y_i - \mathbf{x}_i^T \mathbf{w}^* \quad \text{for any index } i \text{ such that } 0 < \alpha_i^* < C \\ \xi^* &= \max \{ \mathbf{0}_M, \mathbf{1}_M - Y(X\mathbf{w}^* + b^*\mathbf{1}_M) \}\end{aligned}$$

Proof of Proposition 1 From the KKT conditions, the optimal weight vector $\mathbf{w}^\star$ is given by:

$$\mathbf{w}^\star = X^\top Y \boldsymbol{\alpha}^\star.$$

To determine $b^\star$, we use the complementary slackness condition. For any support vector with $0 < \alpha_i^\star < C$, the corresponding constraint is active, yielding:

$$Y_{ii}(X_i \mathbf{w}^\star + b^\star) = 1 \quad \Rightarrow \quad b^\star = Y_{ii} - X_i \mathbf{w}^\star$$

which holds for any i where $0 < \alpha_i^\star < C$. For the slack variables $\boldsymbol{\xi}^\star$, we use the primal constraints:

$$\begin{cases} \mathbf{1}_M - Y(X\mathbf{w}^\star + b^\star \mathbf{1}_M) - \boldsymbol{\xi}^\star \leq \mathbf{0}_M \\ -\boldsymbol{\xi}^\star \leq \mathbf{0}_M \end{cases} \quad \Rightarrow \quad \boldsymbol{\xi}^\star = \max\{\mathbf{0}_M, \mathbf{1}_M - Y(X\mathbf{w}^\star + b^\star \mathbf{1}_M)\}$$

Thus, the primal solution $(\mathbf{w}^\star, b^\star, \boldsymbol{\xi}^\star)$ can be recovered from the dual solution $\boldsymbol{\alpha}^\star$ using the KKT conditions. $\square$

2.2.2 Existence and Uniqueness of the Dual Solution

The dual problem is a convex quadratic problem with linear constraints. The existence and uniqueness of the solution depend on the properties of the Gram matrix G and the constraints imposed by the labels Y .

Theorem 3. Existence & Uniqueness of Dual Solution

The dual problem (DP) always admits at least one global minimizer. Moreover, if YGY is strictly positive-definite on the subspace $\{\boldsymbol{\alpha} \mid \langle \mathbf{y}, \boldsymbol{\alpha} \rangle = 0\}$, then the solution $\boldsymbol{\alpha}^\star$ is unique. In degenerate cases, multiple optimal solutions may exist.

Proof of Theorem 3

Existence of a solution: The dual problem (DP) is a convex quadratic problem in $\boldsymbol{\alpha}$, subject to a linear equality constraint $\boldsymbol{\alpha}^\top Y = \mathbf{0}^\top$ and box constraints $0 \leq \boldsymbol{\alpha} \leq C\mathbf{1}_M$. We define the feasible region as:

$$\Omega = \{\boldsymbol{\alpha} \in \mathbb{R}^M \mid 0 \leq \boldsymbol{\alpha} \leq C\mathbf{1}_M, \boldsymbol{\alpha}^\top Y = \mathbf{0}^\top\},$$

which is the intersection of the closed box $[0, C]^M$ with a hyperplane. This intersection is nonempty: because the dataset includes both positive and negative labels (i.e., $\exists p, n$ such that $y_p = 1, y_n = -1$), one can construct a feasible point by choosing $\alpha_p = \alpha_n$ and setting all other components to zero. Hence, Ω is closed, bounded, and nonempty. The dual objective function, $\tilde{f}(\boldsymbol{\alpha})$, is continuous and convex, being a quadratic form with symmetric matrix YGY and linear term $-\mathbf{1}_M^\top \boldsymbol{\alpha}$. Since f is continuous and convex over a compact feasible set Ω , the extreme value theorem guarantees the existence of a global minimizer $\boldsymbol{\alpha}^\star$.

Uniqueness of a solution: The Hessian of f is $\nabla^2 f(\boldsymbol{\alpha}) = YGY$, which is symmetric and positive semi-definite because G is a Gram matrix and Y is a diagonal matrix with ± 1 entries. If YGY is strictly positive-definite on the subspace $\{\boldsymbol{\alpha} \in \mathbb{R}^M : \boldsymbol{\alpha}^\top Y = \mathbf{0}^\top\}$, then f is strictly convex on the feasible region Ω , and the minimizer $\boldsymbol{\alpha}^\star$ is unique.

In contrast, if YGY is only positive semi-definite (e.g., if G is rank-deficient), then flat directions may exist in the objective, and multiple global minimizers may be possible.

Degenerate cases: Degenerate cases occur when YGY is positive semi-definite but not positive-definite on the constraint subspace. This happens in several scenarios:

1. When the data vectors $\{\mathbf{x}_i\}$ are linearly dependent, making the Gram matrix G rank-deficient. For example, if two feature vectors are identical but have the same label, or if one feature vector is a linear combination of others.
2. When the number of data points M exceeds the dimension N of the feature space, necessarily creating linear dependencies.

In these cases, there exist non-zero directions $\mathbf{v}$ satisfying $\mathbf{v}^\top YGY\mathbf{v} = 0$ and $\mathbf{v}^\top Y = 0$. Moving along such directions from one optimal solution leads to another optimal solution, creating a flat region (typically a face or edge) of the feasible region where all points are global minimizers. Although $\mathbf{w}^\star$ remains unique due to the form $\mathbf{w}^\star = X^\top Y\boldsymbol{\alpha}^\star$, the values of $\boldsymbol{\alpha}^\star$ corresponding to this unique $\mathbf{w}^\star$ are not unique.

3 Numerical Methods

3.1 Gradient Descent for Primal SVM

We are now interested in implementing a numerical solution to the primal SVM problem. Since the objective is convex, gradient-based optimization methods can be used to reliably find a global minimum.

To handle non-separable data, we use a soft-margin formulation, which allows some margin violations at a cost controlled by a regularization parameter. This cost is incorporated using a loss function, e.g., the hinge loss, which penalizes misclassified points. However the hinge loss function is non-smooth, so we instead use a smooth approximation, specifically the logistic loss:

$$\ell(t) = \log(1 + e^{-t})$$

which penalizes misclassified points, and is differentiable. The gradient is computed from the contributions of all training points, weighted by how much they violate the margin, and scaled by the regularization term. Using this gradient, we update the weights iteratively in the direction of steepest descent until convergence to an optimal solution.

3.2 Projected Gradient Descent for Dual SVM

We are now interested in implementing a solution to the dual problem numerically. The reason we are interested in the dual problem numerically is that it allows us to use different kernels, to classify non-linearly which may give us a more accurate classification. We solved the dual SVM problem using projected gradient descent, where each update projects the gradient step back onto the feasible set defined by the constraints. The projection was implemented by converting the problem into a constrained minimization problem given by:

$$\pi_\Omega(\boldsymbol{\beta}) := \min_{\boldsymbol{\alpha} \in \mathbb{R}^M} \frac{1}{2} \|\boldsymbol{\alpha} - \boldsymbol{\beta}\|_2^2 \quad \text{s.t.} \quad \begin{cases} \mathbf{y}^\top \boldsymbol{\alpha} = 0 \\ \mathbf{0}_M \leq \boldsymbol{\alpha} \leq C\mathbf{1}_M \end{cases}$$

This way we can use the Lagrangian method with one lagrange multiplier, λ , and solve the projection efficiently via bisection.

3.2.1 Step Length Selection

To determine appropriate step lengths in both gradient descent methods, we implemented the *Barzilai-Borwein (BB) method*, which dynamically estimates optimal step sizes based on information from previous iterations. When the BB method produced unusable step lengths (e.g., negative or excessively large values), we defaulted to a conservative fixed step size to ensure algorithm stability. In some cases where the step length was not satisfactory it would be necessary to perform a line search. This does come at a higher computational cost. So to avoid unnecessary line searches, we tracked a reference function value, f_{ref} , and only triggered a line search when the objective increased beyond this threshold. If the function hadn't improved after a set number of steps, we updated the reference and potentially performed a line search. This approach gave fast convergence while keeping computational cost low.

4 Experiments and Results

In this section, we present the experimental validation of our SVM implementations. We start with a synthetic linearly separable dataset to compare the primal and dual formulations, followed by experiments with non-linear data using various kernels, and finally evaluate performance on the real-world Breast Cancer Wisconsin dataset[WM91].

4.1 Linear Separable Data

First we generate a linearly separable synthetic dataset with 100 points in 2D using `make_toy_data` function to create data with a clear margin between the two classes. Our optimization problem has the solutions $\mathbf{w}^* = [1, 1]$, and $b^* = 0$ with noise parameter $\sigma = 0.1$ used in data generation.

4.1.1 Primal SVM

We trained the Primal SVM with $C = 1.0$ using subgradient descent for 10,000 iterations with a learning rate (lr) of 0.001 and Barzilai-Borwein steps. Figure 1 shows the decision boundary, margins, and support vectors identified by the model.

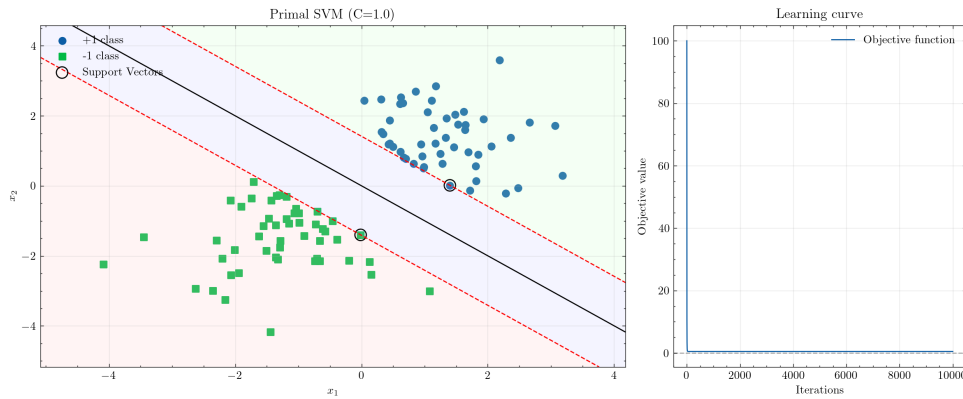


Figure 1: Primal SVM with linear kernel ($C = 1.0$). The solid line represents the decision boundary, and the dashed lines represent the margins. Support vectors are highlighted with black circles.

The primal formulation achieved almost $\approx 100\%$ accuracy on the training data. The algorithm correctly identified two support vectors (one from each class), which lie exactly on the margin

boundaries. The weight vector converged approximately to $\mathbf{w} \approx [0.7069, 0.7073]$, which is just the normalized primal solution. with a bias term $b = -0.001$, resulting in a maximum margin hyperplane.

4.1.2 Dual SVM with Linear Kernel

We trained the Dual SVM with the same $C = 1.0$ parameter using a linear kernel, and achieved 100% accuracy, and identified the same support vectors as the primal approach. When we recovered the weight vector from the dual solution we obtained $\mathbf{w} \approx [0.7071, 0.7071]$. This confirms the theoretical equivalence of the primal and dual formulations for linear separable data.

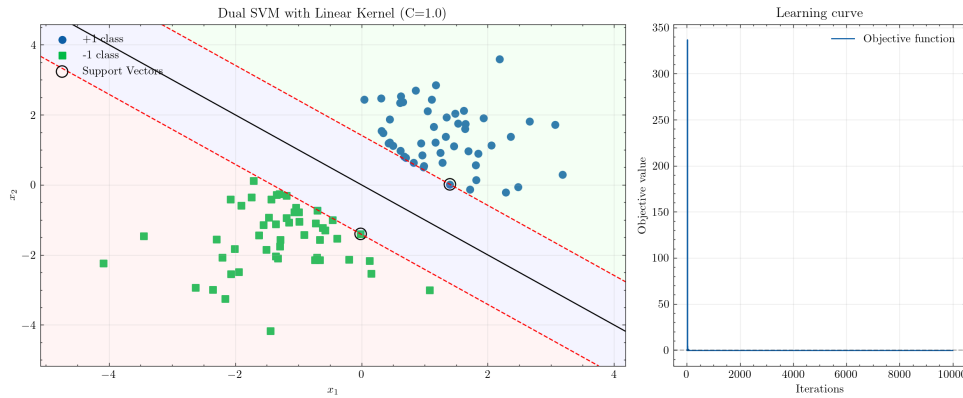


Figure 2: Dual SVM with linear kernel ($C = 1.0$). The decision boundary and margins are nearly identical to those found by the primal formulation.

4.1.3 Dual SVM with RBF Kernel

We trained the Dual SVM with an RBF kernel ($C = 1.0, \gamma = 0.5$) on the same dataset. The model again achieved 100% accuracy. The RBF kernel allows the model to create a non-linear decision boundary, which perfectly separates the two classes with 100% accuracy. The support vectors are again highlighted, and we observe that they are not necessarily on the margin boundaries as in the linear case.

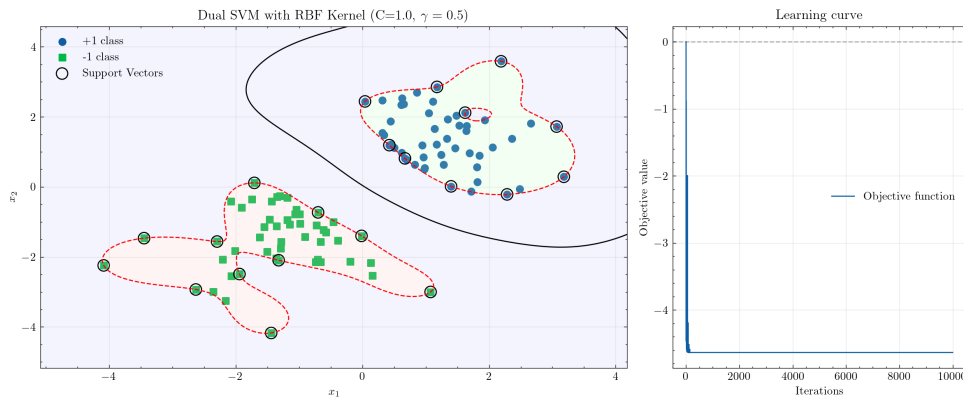


Figure 3: Dual SVM with RBF kernel ($C = 1.0, \gamma = 0.5$). The decision boundary is non-linear, effectively separating the two classes.

4.2 Effect of Regularization Parameter

To understand the impact of the regularization parameter C , we trained the primal SVM with three different values: $C = \{0.1, 1.0, 10.0\}$. Figure 4 shows how the decision boundaries and margins change with different C values.

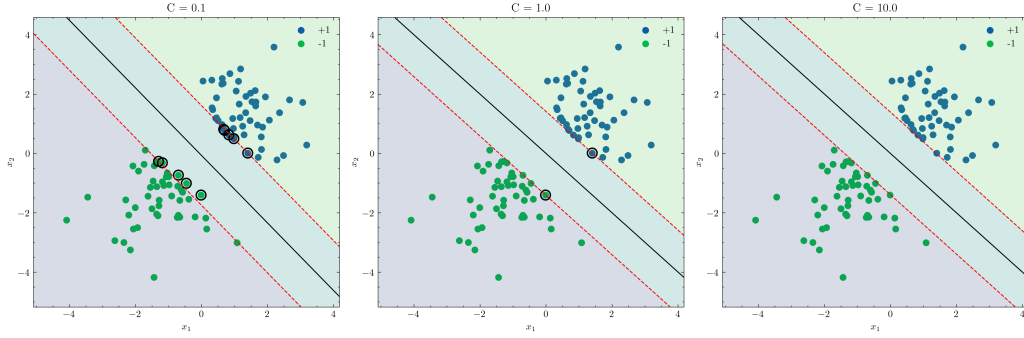


Figure 4: Comparison of different C values in Primal SVM. Lower C values (left) result in wider margins but may allow more training errors, while higher C values (right) enforce stricter classification at the cost of narrower margins.

- With $C = 0.1$ (low regularization), the model prioritizes a wider margin at the potential cost of misclassifications.
- With $C = 1.0$ (balanced), we achieve a good trade-off between margin width and classification accuracy.
- With $C = 10.0$ (high regularization), the model focuses more on reducing training errors, potentially leading to narrower margins and possible overfitting.

4.3 Non-linear Data with Different Kernels

We next evaluate our dual SVM implementation on non-linearly separable data using different kernels. Figure 5 shows the performance of four different kernels on a synthetic dataset with circular decision boundaries.

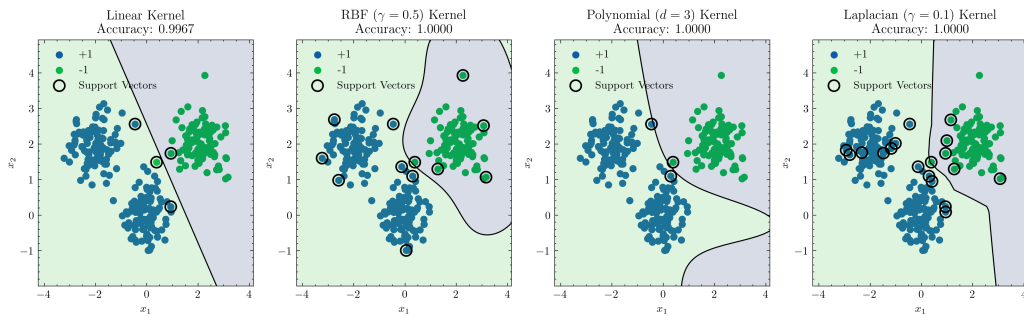


Figure 5: Comparison of different kernels on non-linear data. From left to right: Linear, RBF, Polynomial, and Laplacian kernels.

As expected, the linear kernel performs the “worst” on the non-linear dataset, with a 99.67% accuracy. The RBF kernel with $\gamma = 0.5$ perfectly captures the circular decision boundary with 100% accuracy. The polynomial kernel (degree 3) also performs well with 100% accuracy but creates

a more complex decision boundary. The Laplacian kernel ($\gamma = 0.1$) also achieves 100% accuracy with a smooth decision boundary. This demonstrates the flexibility of SVMs with different kernel functions, allowing them to adapt to various data distributions.

4.4 Convergence Analysis of Different Kernels

To examine kernel-specific optimization, we tracked dual SVM convergence on the non-linear dataset (Fig. 6). RBF converged fastest (80 iterations), linear took 100 with mild oscillations, Laplacian needed 200 stable steps, and polynomial exhibited the most instability, converging after 200. This aligns with theory that RBF is usually better-conditioned, while polynomial kernels may be numerically challenging[Zha01].

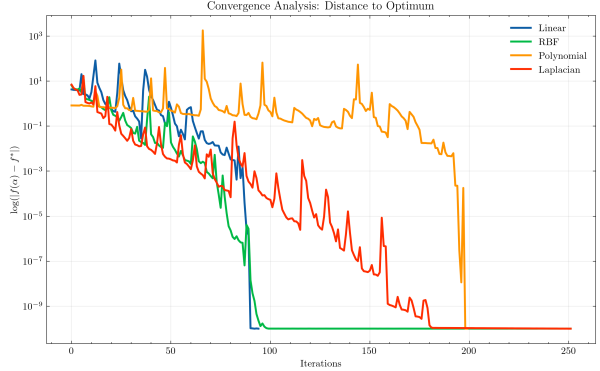


Figure 6: Convergence behavior of different kernels during optimization.

4.5 Breast Cancer Wisconsin Dataset

Finally, we applied our SVM implementation to the Breast Cancer Wisconsin dataset, which contains 30 features computed from digitized images of fine needle aspirates (FNA) of breast mass, with binary labels indicating benign or malignant tumors.

4.5.1 Kernel Selection and Hyperparameter Tuning

We evaluated multiple kernel types (Linear, RBF, Polynomial, Laplacian) with different C values (0.1, 1.0, 10.0) on this dataset. Table 2 shows the results.

Kernel	$C = 0.1$		$C = 1.0$		$C = 10.0$	
	Train	Test	Train	Test	Train	Test
Linear	0.9824	0.9825	0.9868	0.9561	0.5077	0.5175
RBF ($\gamma = 0.01$)	0.9516	0.9649	0.9758	0.9649	0.9912	0.9825
Polynomial (d=2)	-	-	0.7780	0.7719	-	-
Laplacian ($\gamma = 0.1$)	-	-	0.9934	0.9649	-	-

Table 2: Performance of different SVM configurations on the Breast Cancer Wisconsin dataset.

The RBF kernel with $C = 10.0$ achieved the best test accuracy of 98.25%. Interestingly, the linear kernel with $C = 0.1$ performed nearly as well, suggesting that the data might be largely linearly separable in the original feature space. The dramatic drop in performance for the linear kernel with $C = 10.0$ indicates possible numerical issues or instability with high regularization values.

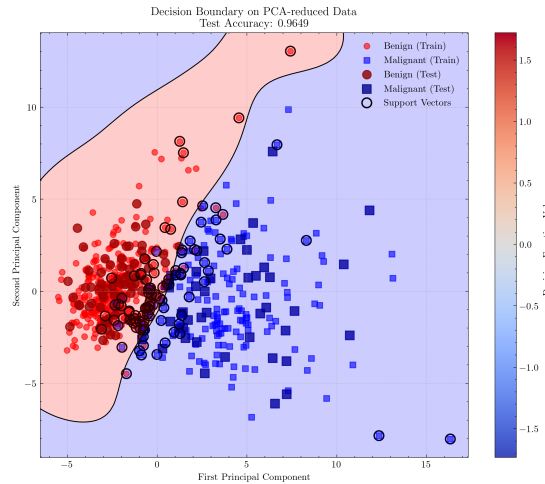


Figure 7: Decision boundary visualization of the Breast Cancer dataset after PCA reduction. Red points represent benign tumors, blue points represent malignant tumors, and black circles indicate support vectors.

4.5.2 Discussion

Our experiments showed that primal and dual SVMs yield identical solutions for linearly separable data. The parameter C strikes a balance between margin width and misclassifications, thus impacting generalization. Non-linear kernels greatly improve performance on data sets with complex decision boundaries. When applied to real data like Breast Cancer, the RBF kernel generally performs best, though the linear kernel also worked well. Convergence relies on careful tuning of learning rates, tolerance thresholds, and iterations. These findings illustrate SVMs’ flexibility and effectiveness, especially when paired with suitable kernels for non-linear tasks.

5 Conclusion

This project investigated Support Vector Machines theoretically, numerically, and practically. We formulated soft-margin SVMs in both primal form and dual form (enabling kernels), proved solution existence and uniqueness, and verified their equivalence through KKT conditions. Our numerical implementations included gradient descent with smooth approximations, projected gradient descent, adaptive step sizes, and line search techniques. Experiments revealed the critical role of the regularization parameter C , different convergence behaviors across kernels, and superior RBF kernel performance (98.25% accuracy) on the Breast Cancer Wisconsin dataset. In conclusion, SVMs remain a valuable tool, offering theoretical rigor, convex optimization benefits, interpretable decision boundaries, and remarkable efficiency when working with smaller datasets.

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